MIL-B-28514A(YD)
31 May 1983
SUPERSEDING
MIL-B-28514(YD)
4 January 1968

MILITARY SPECIFICATION

BOOTH, SPRAYING, PAINT (WITH EXHAUST SYSTEM)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers one type of dry baffle, floor-type paint spraying booth with exhaust system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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MIL-E-16298 - Electric Machines, Having Rotating Parts and Associated Repair Parts; Packaging of.
MIL-E-17555 - Electronics and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The paint spraying booth shall consist of a cabinet, baffles, lighting fixtures, exhaust fan with motor, exhaust pipe, weather canopy, roof flange, light switch, electrical connections, and other components necessary to form a complete paint spraying booth.

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3.2 Safety standard. Electrical wiring and associated electrical components shall conform to and be installed in accordance with NFPA No. 70.

3.2.1 Certification. Evidence of meeting the applicable NFPA standards shall be the contractor's certified statement that the paint booth conforms to all applicable NFPA standards.

* 3.3 Standard commercial product. The paint spraying booth shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the paint spraying booth being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.5 Identical items. All paint spraying booths of the same classification, furnished under any specific contract, shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

3.6 Design. The paint spraying booth shall be a dry baffle-type floor-mounted booth, designed for complete removal of fumes and vapors from the spray area through a top outlet exhaust pipe. The booth shall be designed for intermittent or light duty service.

* 3.6.1 Cabinet. The cabinet shall be three wall, riveted- or bolted-panel construction with panels and joints reinforced to make the cabinet self-supporting. The panels shall be fabricated from not less than 0.0478 (US revised standard gage No. 18) galvanized sheet steel. The panel edges shall be rolled or reinforced and the corners mitered and welded. All bolts and fittings necessary to assemble the cabinet shall be furnished.

* 3.6.2 Baffles. Baffles shall be fabricated from not less than 0.0478 (US revised standard gage No. 18) sheet steel positioned to provide maximum working area and arranged to assure uniform air distribution throughout the booth. The baffles shall be removable for cleaning.

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* 3.6.3 Lighting fixtures. Unless otherwise specified (see 6.2), the work area of the booth shall be illuminated, by either incandescent or fluorescent fixtures. Lighting fixtures shall be top-mounted and vaporproof with conduit connections. If of an incandescent type, three fixtures suitable for use with at least 150-watt lamps shall be provided. If of a fluroescent type, fixtures shall be suitable for use with 40-watt lamps and of sufficient number to provide, as a minimum, the same degree of illumination as three 150-watt incandescent lamps.

* 3.6.4 Exhaust fan and motor. The exhaust fan shall be capable of discharging a volume of air in cubic feet per minute (cfm) at 1/4-inch static pressure equivalent to the product of the spray booths working face area (width times height in feet) and the rate of desired air flow through the booth, the latter to be taken as not less than 125 cfm unless otherwise specified (see 6.2). The fan blade speed and pitch shall be such as to insure that the fan blade tip speed does not result in a noise level in excess of that permitted by applicable OSHA regulations in effect at the time of manufacture. The fan shall consist of a nonsparking or nonferrous blade and shall be properly sized to the exhaust pipe (see 3.6.5), and connected to a shaft mounted on self-alining, permanently lubricated ball bearings. The exhaust fan motor shall be a totally enclosed fan cooled type designed to operate on a nominal 208 volts, 60 Hertz (Hz), 3-phase alternating current (ac) in a hazardous location, class 1, group D, as defined by NFPA No. 70. Pulleys and belts provided shall be adjustable for belt tension and alinement. Belting shall be of the V-belt, endless type and shall be installed for ease of replacement without dismantling any integral parts of the components to which attached.

* 3.6.5 Exhaust pipe. The exhaust pipe shall consist of four 24-inch diameter by 36-inch long, plain, straight, 0.0359 (US revised standard gage No. 20) galvanized metal sections. The sections shall have a lock seam and one end reduced for easy installation. An exhaust pipe fan connecting ring shall be furnished. One length of exhaust pipe shall include a cleanout door and non-motorized automatic damper.

* 3.6.6 Weather canopy and bird screen. The weather canopy shall be of 0.0359 (US revised standard gage No. 20) galvanized metal and shall have an air diffusion cone to prevent turbulence and back pressure. The canopy shall be provided with guy wire, anchors, turnbuckles, and clamps. A bird screen shall be provided to keep debris, leaves, and birds out of the exhaust stack.

* 3.6.7 Roof flange. The roof flange shall be 24 inches in diameter and fabricated from 0.0359 (US revised standard gage No. 20) galvanized sheet steel. The flange shall be adjustable for installation on a flat roof and roofs having a pitch up to one-half.

3.6.8 Hardware. All nuts, bolts, fittings, and seals necessary to assemble the complete unit shall be provided.

3.7 Dimensions and weight.

3.7.1 Dimensions. The inside (working) dimensions of the booth shall be not less than 7-1/2 feet wide by 7-feet high by 6-feet deep.

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* 3.7.2 Weight. The weight of the paint spraying booth, less the weights of the exhaust fan, motor, and pipe (see 3.6.4 and 3.6.5), the weather canopy, bird screen, and roof flange (see 3.6.6 and 3.6.7), but including the baffles and lighting fixtures (see 3.6.2 and 3.6.3), shall not exceed 1,500 pounds.

3.8 Electrical wiring and components. Except for the exhaust fan motor, the electrical components shall be designed to operate on 110 volts, 60 Hz, single-phase ac. All wiring shall be color coded and shall conform to the requirements of NFPA No. 70. Switches for lights and fan motor shall be mounted, prewired, and shipped knocked-down.

3.8.1 Light switch. An on-off explosion-proof light switch to control the overhead lights shall be furnished.

3.8.2 Magnetic starter. The booth shall be provided with a magnetic starter of the automatic type suitable for starting the fan motor (see 3.6.4).

* 3.8.3 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.9 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A or C. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

* 3.10 Identification marking. Identification shall be permanently and legibly marked directly on the paint spraying booth or on a corrosion-resisting metal plate securely attached to the paint spraying booth at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer. The nomenclature shall be "BOOTH, SPRAYING, PAINT."

3.10.1 Instruction plates. The paint spraying booth shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws, bolts, or rivets of not less than 1/8-inch diameter.

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3.11 Workmanship.

* 3.11.1 Steel fabrication. Steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.11.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.5 Wiring. All wiring shall be enclosed in metal tubing where possible and in all cases shall be secured and free of scuffing, kinking, or straining.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Examination. Each paint spraying booth shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all

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visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.3 Test. Each complete spray booth shall be given an operational test. The exhaust fan installation shall be visually inspected to insure that there is no binding between the fan and exhaust pipe. The lighting system shall be operated and the exhaust fan motor shall be started and brought up to speed. After 5 minutes the motor shall be shut down and checked for any visible signs of overheating. Failure to pass the test shall be cause for rejection of the unit. Rejected units may be reworked or have parts replaced to correct defects. Before reinspection, full particulars concerning actions taken to correct defects shall be furnished.

4.4 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.2.1. The Government reserves the right to examine and test all spray booths to determine the validity of the certification.

* 4.5 Packaging inspection. The preservation, packaging, packing, and marking of the paint sprayiang booth shall be inspected to verify conformance to the requirements of section 5.

* 5. PACKAGING

* 5.1 Preservation. The preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

* 5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, screw pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.1.1.3 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1.

* 5.1.1.5 Electrical components. Motors, light fixtures, and other electrical controls shall be preserved and packaged in accordance with the level A requirements of MIL-E-17555 and MIL-P-16298.

* 5.1.1.6 Hardware and exhaust pipes. Nuts, bolts, fittings, seals, and exhaust pipes shall be packaged in boxes conforming to PPP-B-636, weather

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resistant. The box shall be closed and sealed in accordance with the appendix to the box specification. Cushioning shall be used within the box to prevent movement of the contents.

* 5.1.1.7 Guy wires. The guy wires shall be coiled to a minimum safe diameter and the coil secured.

* 5.1.1.8 Technical publications. Technical publications for each paint booth shall be preserved method IC-1 or IC-3.

* 5.1.1.9 Consolidated packaging. The packaged hardware, guy wires, technical publications, and other small components shall be consolidated packaged in close-fitting boxes conforming to PPP-B-601, domestic type, or PPP-B-621, class 1. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

* 5.1.2 Commercial. Material shall be packaged in accordance with MIL-STD-1188.

* 5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

* 5.2.1 Level A. Each complete paint booth shall be packed in a crate conforming to MIL-C-104, type II, style a, class optional. The contents shall be blocked, braced, and anchored as applicable, in accordance with the crate specification.

* 5.2.2 Level B. Each complete paint spraying booth shall be packed in a crate conforming to MIL-C-52590, type V, style A; or MIL-C-3774, type I, style A. The contents shall be anchored, blocked, braced, and waterproofed with a shroud in accordance with the crate specification.

5.2.3 Commercial. Material shall be packed in accordance with MIL-STD-1188.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The paint spraying booth covered by this specification is intended to be used for painting of parts in optical and fire control shops.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When either incandescent or fluorescent type lighting fixtures are specifically required (see 3.6.3).
c. When air flow velocity through the spray booth is to be other than 125 cfm (see 3.6.4).
d. When fungus proofing is required (see 3.8.3).
e. When cleaning, treatment, and painting is to be in accordance with MIL-T-704, and the color of the desired finished coat of paint (see 3.9).
f. Level of preservation and level of packing required (see 5.1 and 5.2).

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* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:

Navy - YD

Project No. 4940-N370

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

4940-N370.

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-28514A(YD)	BOOTH, SPRAYING, PAINT (WITH EXHAUST SYSTEM)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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